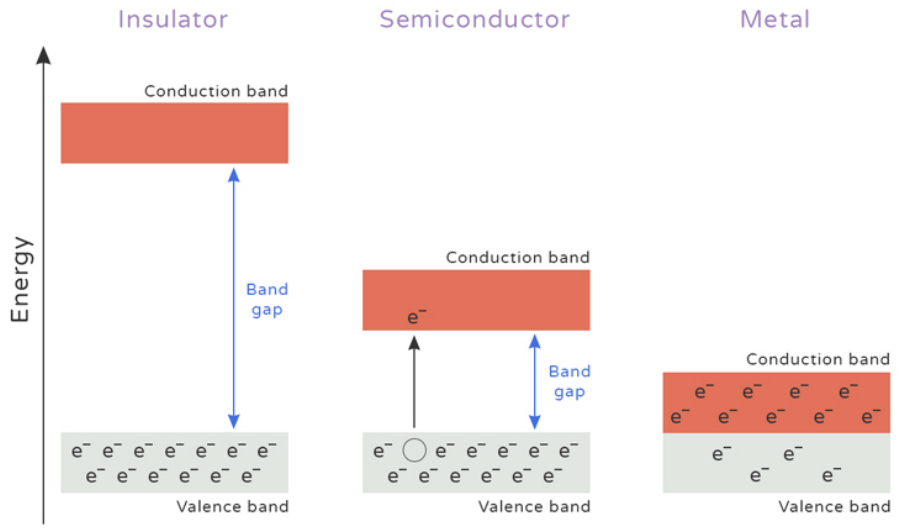
Electronics Basics



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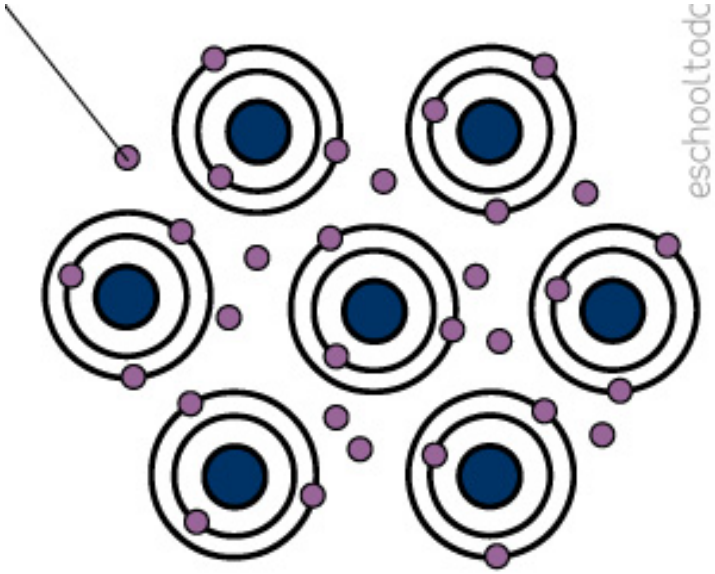
Atomic Structure

- In a single atom, electrons occupy different energy levels or shells.
- When atoms join together in a material, these energy levels form energy bands.
- The valence band contains the outer electrons that are normally bound within the material.
- The conduction band is a higher-energy band where electrons are free to move through the material.
- Between them is the band gap. Electrons must gain enough energy to cross this gap before they can move freely and contribute to electrical current.



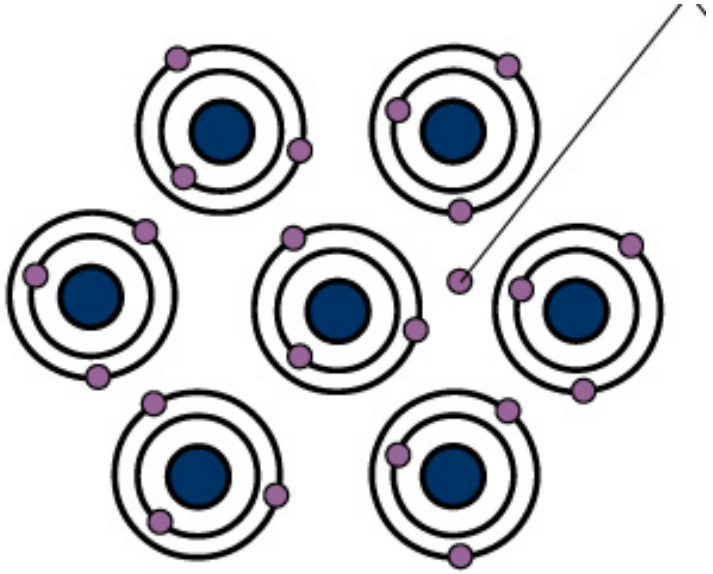
Conductors

- In conductors the valence and conduction bands overlap, so there are many free-floating electrons which can move through the material
- These delocalised electrons form an “electron sea” around the atoms.
- With no applied voltage, the electrons move randomly, so there is no net current.
- When a voltage is applied, the electrons gain a small overall drift in one direction, producing an electric current.



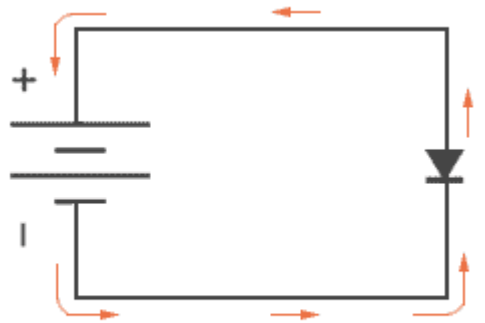
Insulators

- In insulators, electrons are held tightly to their atoms and are not free to move through the material.
- The valence band is full and separated from the conduction band by a large band gap.
- Under normal conditions, electrons cannot gain enough energy to reach the conduction band, so very little current can flow.
- Examples: glass, rubber, plastic, and dry wood.
- Insulators are used to stop current flow and provide safety in circuits.



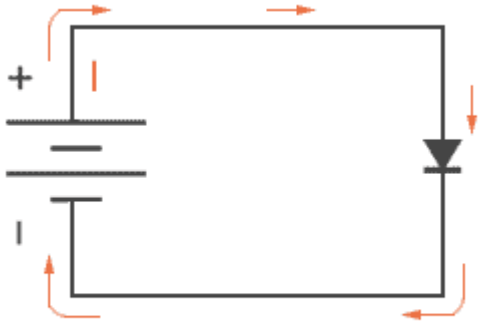
Conventional vs Electron Flow

- **Electron Flow**
  - Actual physical movement of electrons.
  - Electrons drift from negative (–) to positive (+) terminal.
  - Matches what's really happening in the conductor.



Electron Current

- **Conventional Current Flow**
  - Defined historically before electrons were discovered.
  - Current assumed to flow from positive (+) to negative (–).
  - Still used today in circuit diagrams and engineering.



Conventional Current

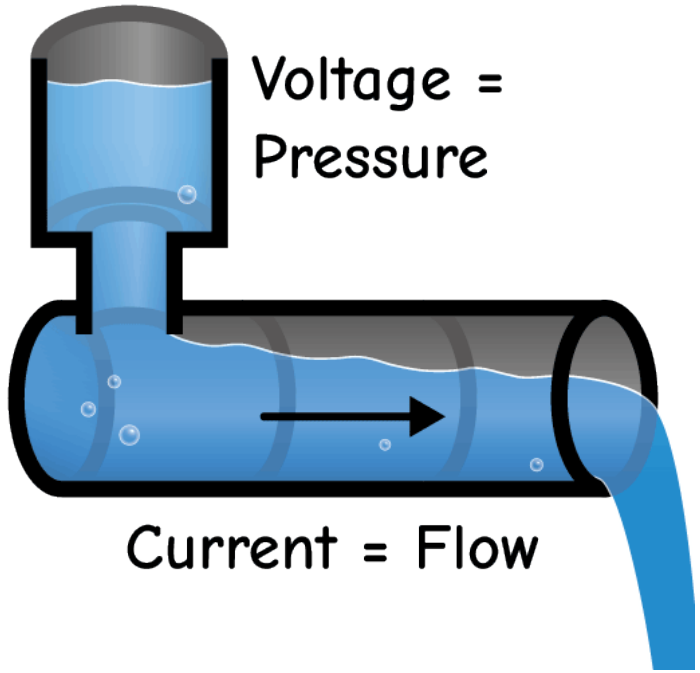
Current

- Current is the rate at which electric charge flows through a circuit.
- Current has the symbol *I* and is measured in amperes (A).
- 1 ampere = 1 coulomb of charge passing a point each second.

Analogy

- Think of current like the flow rate of water through a pipe.
- More water passing a point each second = greater flow rate.
- In a circuit, more charge passing a point each second = greater current.

$$I = \frac{Q}{t}$$



Measuring Current

- We are going to build a simple resistor circuit
- Set the voltage on your power supply to 5 volts
- Connect the multimeter in series and measure the current
- Record it in the table on the worksheet



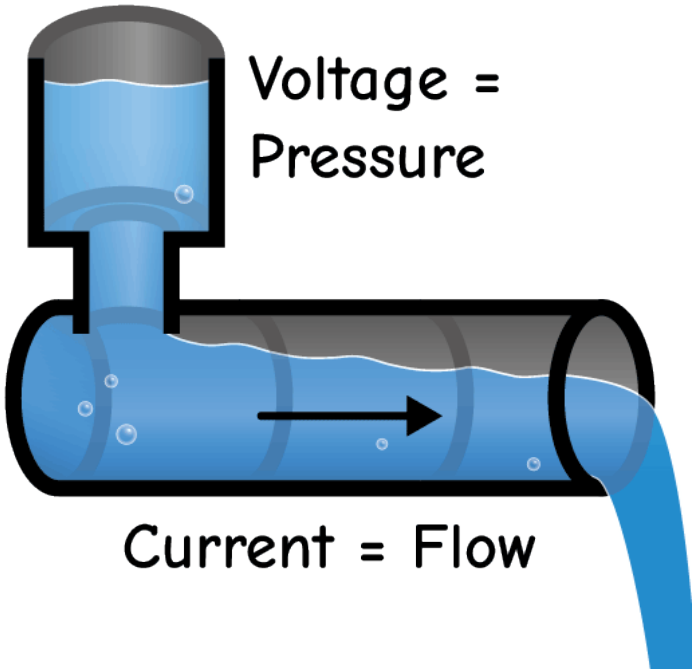
Voltage

- Voltage is the potential difference between two points in a circuit.
- It tells us how much energy is transferred per unit charge.
- Voltage has the symbol *V* and is measured in volts (V).
- 1 volt = 1 joule of energy per coulomb of charge.

Analogy

- Voltage is like the pressure difference pushing water through a pipe.
- Higher voltage means each coulomb of charge can transfer more energy.
- Voltage can be provided by batteries, power supplies and generators.

$$V = \frac{W}{Q}$$



Measuring Voltage

- We are going to build a different simple resistor circuit
- Set the voltage on your power supply to 5 volts
- Connect the multimeter in parallel to the 180Ω resistor and measure the voltage
- Record it in the table on the worksheet

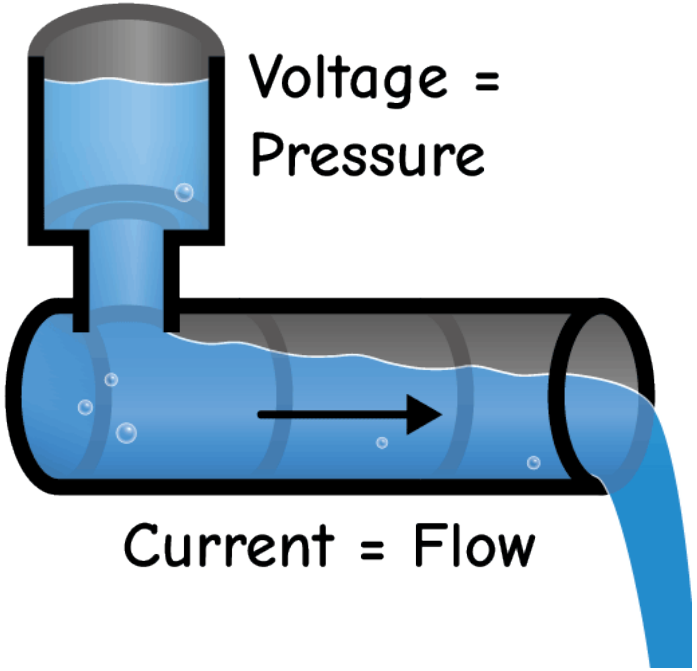


Resistance

- Resistance is the opposition to the flow of electric current.
- Resistance has the symbol *R* and is measured in ohms (Ω).

Analogy

- Resistance is like a restriction in a water pipe.
- A greater restriction makes it harder for water to flow.
- Similarly, greater electrical resistance makes it harder for current to flow.



Measuring Resistance

- We are going to attach a single resistor to our boards
- Disconnect our power supplies
- Connect the multimeter in parallel to the resistor and measure the resistance
- Record it in the table on the worksheet

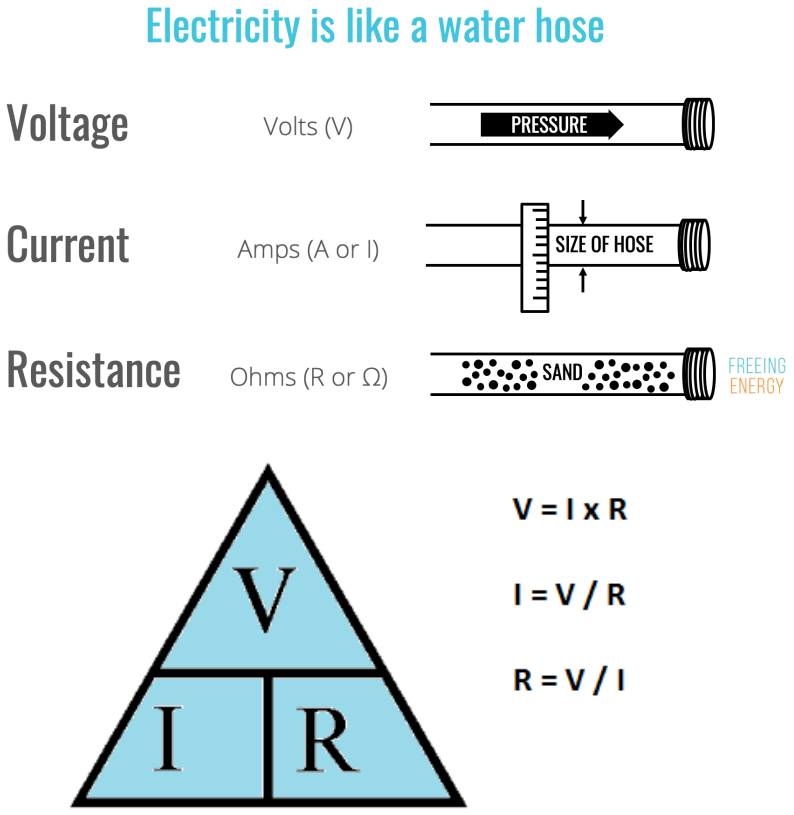


V = IR

- Ohm's Law describes the relationship between voltage, current and resistance in a circuit.
- For a resistor, increasing the voltage increases the current, while increasing the resistance decreases the current.
- The relationship is:

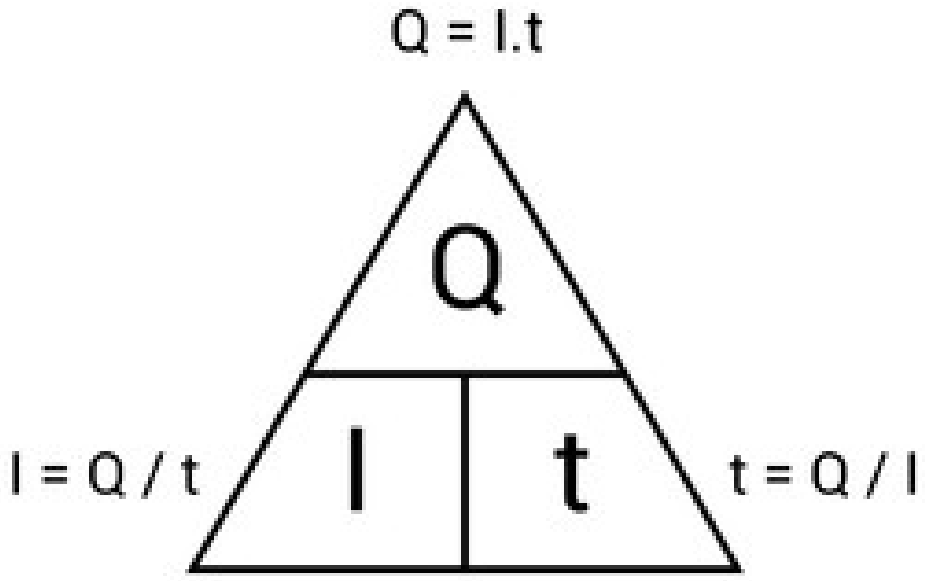
$$V = IR$$

Where:
V = voltage in volts
I = current in amperes
R = Resistance in ohms



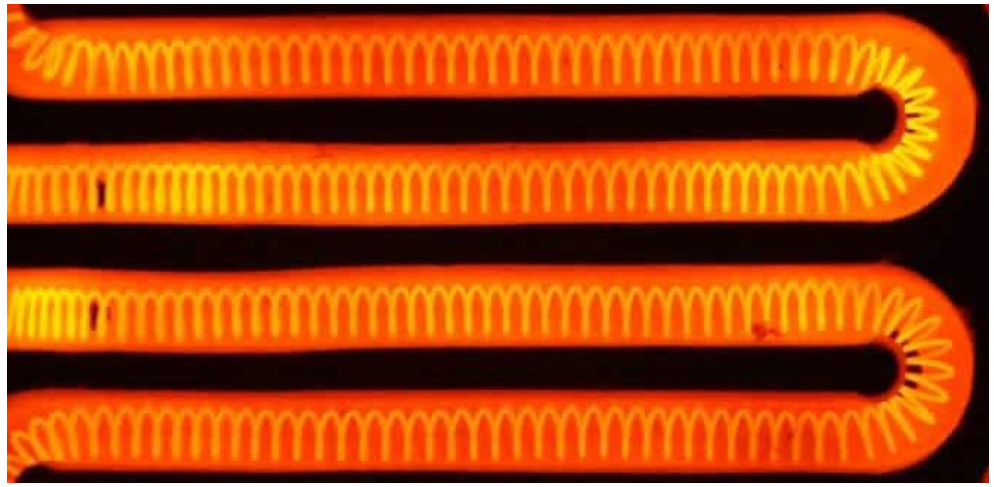
Charge

- Electric charge is a property of particles that causes them to experience electrical forces.
- There are two types of charge: positive (+) and negative (−).
- Protons have positive charge and electrons have negative charge.
- Charge, *Q*, is measured in coulombs (C).
- Electron charge is $-1.6 * 10^{-19}$ C
- $Q = I * t$ (charge = current * time)



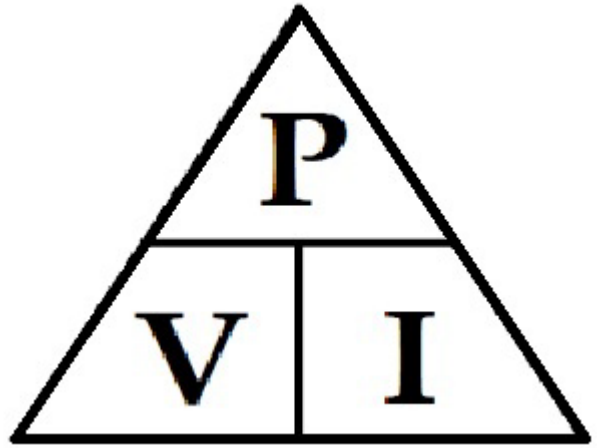
Energy

- Energy is the ability to do work or cause change.
- Electrical energy has the symbol *E* and is measured in joules (J).
- Electrical energy transferred depends on voltage, current and time.
- It has the formula: $E = V \times I \times t$
- Where:
  - V = Voltage (volts)
  - I = Current (amps)
  - t = Time (seconds)
- Examples:
  - A 60W bulb lit for 1 hour uses 216,000J.
  - Phone battery stores energy in the range of 40,000 to 70,000 J.



Power

- Power is the rate at which energy is transferred.
- It has the symbol *P* and the units Watts (W)
- 1 watt = 1 joule of energy transferred per second.
- Electrical power depends on both voltage and current.
- A device with a higher power rating transfers more energy each second.
- Examples:
  - Phone charger ≈ 10 to 30 W
  - Kettle ≈ 2000 to 3000 W
  - Light bulb ≈ 5 to 60 W



$$P = IV$$

Electric Force

- Electric force is the force of attraction or repulsion between charged particles.
- Opposite charges attract. Like charges repel.
- The force becomes stronger when:
  - the charges are larger
  - the charges are closer together
- In a circuit, an electric field exerts a force on electrons and causes them to drift through the conductor.
- Voltage creates an electric field; the electric field exerts a force on charge.

